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Tutorial 12

1. $X \sim B(10, 0.12)$ (Binomial Distribution)

a)

$$P(X=3) = \binom{n}{x} \times p^x \times (1-p)^{n-x}$$

$$= \binom{10}{3} \times 0.12^3 \times (1-0.12)^{10-3}$$

$$\approx 0.0847$$

b) $$P(X=6) = \binom{n}{x} \times p^x \times (1-p)^{n-x}$$

$$= \binom{10}{6} \times 0.12^6 \times (1-0.12)^{10-6}$$

$$\approx 0.3760 \times 10^{-3}$$

c) $P(X \leq 2)$

$$F(2) = P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= \binom{10}{0} \times 0.12^0 \times (1-0.12)^{10}$$

$$+ \binom{10}{1} \times 0.12^1 \times (1-0.12)^{10-1}$$

$$+ \binom{10}{2} \times 0.12^2 \times (1-0.12)^{10-2}$$

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$\approx 0.8913$

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d) $P(X \geq 7)$

$P(X \geq 7) = P(X=7) + P(X=8) + P(X=9) + P(X=10)$

$= \binom{10}{7} \times 0.12^7 \times (1-0.12)^{10-7}$

$+ \binom{10}{8} \times 0.12^8 \times (1-0.12)^{10-8}$

$+ \binom{10}{9} \times 0.12^9 \times (1-0.12)^{10-9}$

$+ \binom{10}{10} \times 0.12^{10} \times (1-0.12)^{10-10}$

$\approx 3.0846 \times 10^{-5}$

e) The expectation is:

$E(X) = np = 0.12 \times 10$

$= 1.2$

f) The variance is:

$Var(X) = n \times p \times (1-p)$

$= 10 \times 0.12 \times (1-0.12)$

$= 1.056$

2.

- $B(6, 0.5)$

We have tabular form:

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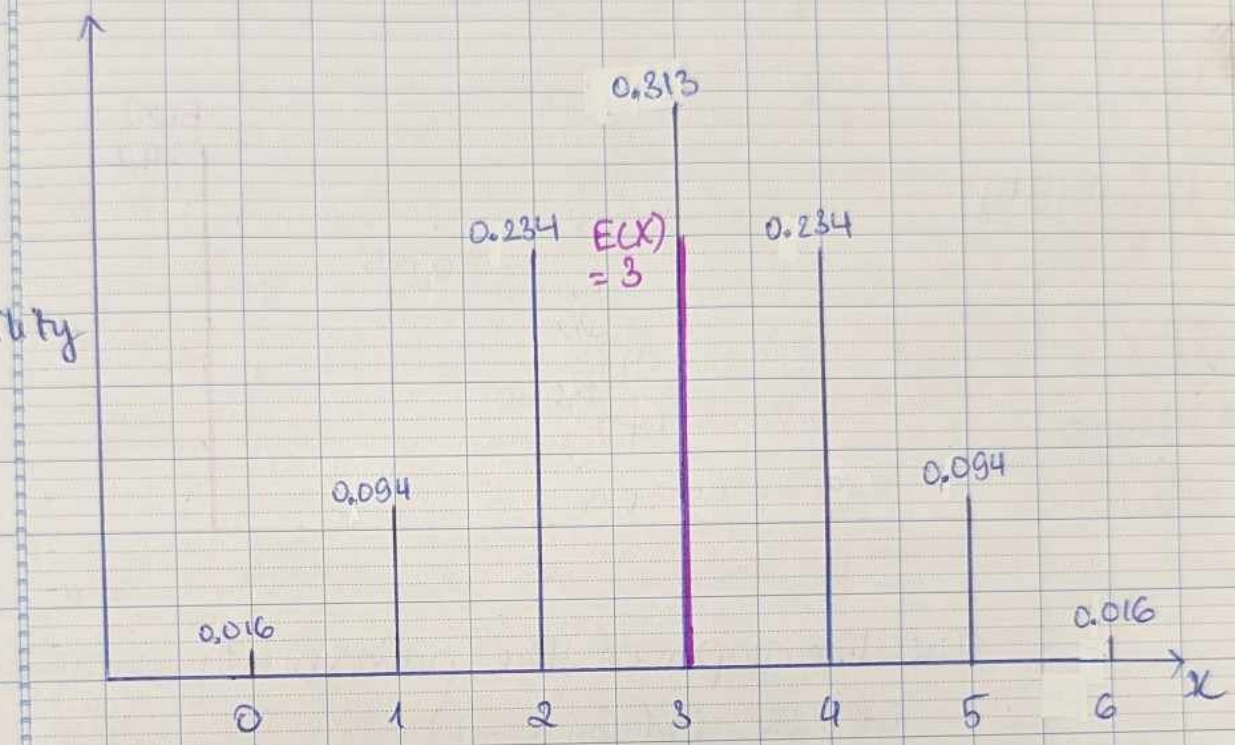
| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $P(X=x)$ | $1/64$ | $3/32$ | $15/64$ | $5/16$ | $15/64$ | $3/32$ | $1/64$ |




The line graph of the probability mass function of a $B(6, 0.5)$

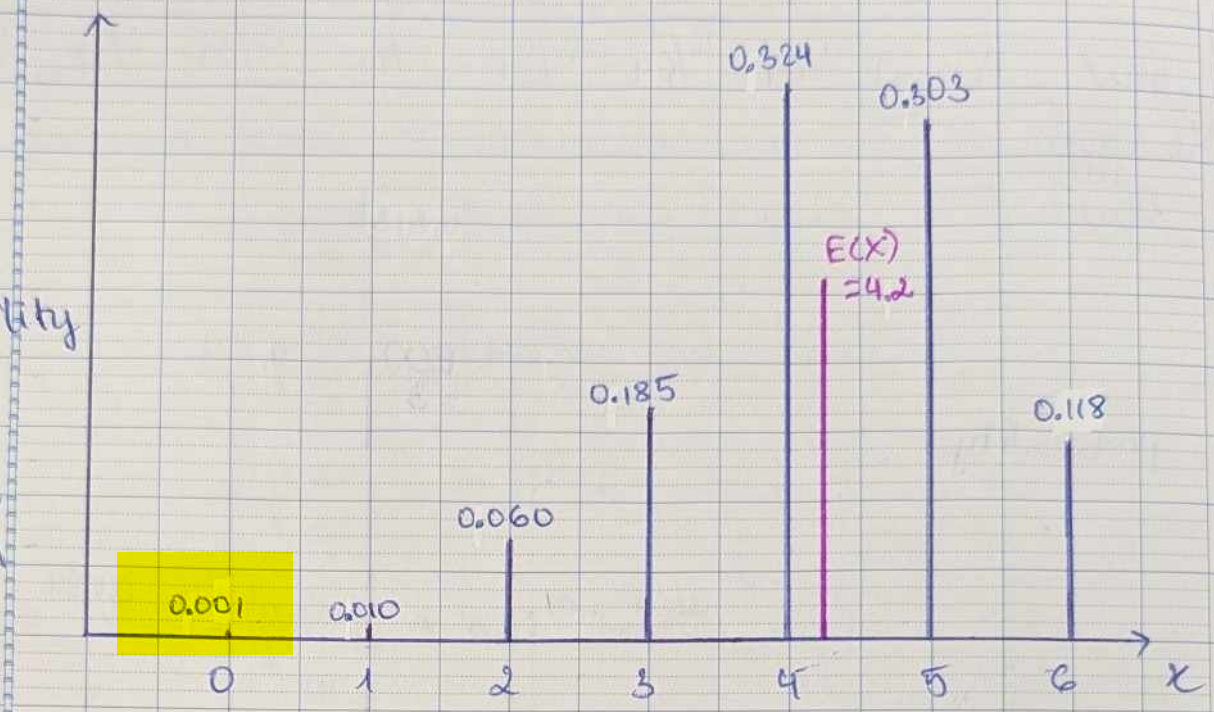
The standard deviation is:

$$\sigma = \sqrt{Var(X)} = \sqrt{n \times p \times (1-p)}$$
$$= \sqrt{6 \times 0.5 \times (1-0.5)}$$
$$= \frac{\sqrt{6}}{2} \approx 1,2247$$

- $B(6, 0.7)$

We have tabular form:

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $P(X=x)$ | $7.29 \times 10^{-4}$ | 0.010 | 0.060 | 0.185 | 0.324 | 0.303 | 0.118 |

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The line graph of the probability mass function of a $B(6, 0.7)$

The standard deviation is:

$$\sigma = \sqrt{Var(X)} = \sqrt{n \times p \times (1-p)}$$
$$= \sqrt{6 \times 0.7 \times (1-0.7)}$$
$$= \frac{3\sqrt{14}}{10} \approx 1.122$$

3.

a) The probability of rolling even number:

$P(\text{even}) = \frac{1}{2}$

Let X be the number of even numbers obtained. Suppose that a fair coin is tossed 8 times.

Then $X \sim B(8, 0.5)$

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The probability is:

$$P(X=5) = \binom{8}{5} \times 0.5^5 \times (1-0.5)^{8-5}$$

$$\approx 0.21875$$

b) The probability of rolling 6:

$$P(6) = \frac{1}{6}$$

Let X be the number of 6s obtained. Suppose that a fair coin is tossed 8 times:

Then $X \sim B(8, 1/6)$

The probability is:

$$P(X=1) = \binom{8}{1} \times \left(\frac{1}{6}\right)^1 \times \left(1-\frac{1}{6}\right)^{8-1}$$

$$\approx 0.3721$$

c) The probability of rolling one 4:

$$P(4) = \frac{1}{6}$$

Let X be the number of 4s obtained. Suppose that a fair coin is tossed 8 times.

Then $X \sim B(8, 1/6)$

The probability is:

$$P(X=0) = \binom{8}{0} \times \left(\frac{1}{6}\right)^0 \times \left(\frac{5}{6}\right)^8$$

$$\approx 0,2326$$

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4) The binomial random variable $X_1$ and $X_2$ are the numbers of successes after n1 and n2 trials respectively.

Therefore, the sum y of the two binomial random variables $X_1, X_2$ can be thought of as the sum of the number of successes after n1 trials and n2 trials.

And because p is the same for both $X_1$ and $X_2$, we can apply the formula of binomial random variable to y as:

$$y \sim B(n1 + n2, p)$$

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5) Based on question, X has parameter $p = 0.7$

a) $P(X=4) = (1-p)^{x-1} \times p$

$= (1-0.7)^{4-1} \times 0.7$

$= 0.0189$

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b) $P(X=1) = (1-p)^{x-1} \times p$

$= (1-0.7)^{1-1} \times 0.7$

$= 0.7$

c) $P(X \leq 5) = 1-(1-p)^x$

$= 1-(1-0.7)^5$

$= 0.99757$

d) $P(X \geq 8) = 1 - P(X \leq 7)$

$= 1-(1-(1-p)^x)$

$= 1-(1-(1-0.7)^7)$

$= 0.2187 \times 10^{-3}$

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6) Because $X_1, \dots, X_r$ are independent random variables, so when we sum up them, the probability is still $p$.

Additionally, geometric distribution indicates the probability that is counted the number of trials performed until the first success occurs.

So when we sum up $X_1, \dots, X_r$, that means we calculate the total probability of $r$ success.

In conclusion, we can use a negative binomial distribution with parameters $p$ and $r$ in this case.

The relationship between geometric distribution and negative binomial random variable $X$ can be

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considered as the total of r independent random variable.

So, the mean of geometric distribution is.

$$E(X) = \frac{1}{p}$$

We can know exactly the mean of negative binomial random variable X is: $E(X) = \frac{n}{p}$

Similarly, we have variance in geometric

$$Var(X) = \frac{(1-p)}{p^2}$$

And the variance of the negative binomial random variable is:

$$Var(X) = \frac{r \cdot (1-p)}{p^2}$$

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7)

a) $P(X=5) = \binom{x-1}{r-1} \times (1-p)^{x-r} \times p^{r}$

$= \binom{5-1}{3-1} \times (1-0.6)^{5-3} \times 0.6^{3}$

$= 0.20736$

b) $P(X=8) = \binom{x-1}{r-1} \times (1-p)^{x-r} \times p^{r}$

$= \binom{8-1}{3-1} \times (1-0.6)^{8-3} \times 0.6^{3}$

$\approx 0.0464$

c) $P(X \leq 7) = P(X=3) + P(X=4) + P(X=5)$
$+ P(X=6) + P(X=7)$

$= \binom{3-1}{3-1} \times (1-0.6)^{0} \times 0.6^{3}$

$+ \binom{4-1}{3-1} \times (1-0.6)^{4-3} \times 0.6^{3}$

$+ \binom{5-1}{3-1} \times (1-0.6)^{5-3} \times 0.6^{3}$

$+ \binom{6-1}{3-1} \times (1-0.6)^{6-3} \times 0.6^{3}$

$+ \binom{7-1}{3-1} \times (1-0.6)^{7-3} \times 0.6^{3}$

$\approx 0.9037$

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d) $P(X \geq 7) = 1 - P(X \leq 6)$

$= 0.1792$

• $P(X \leq 6) = P(X = 3) + P(X = 4) + P(X = 5)$
$+ P(X = 6)$

$$= \binom{3-1}{3-1} \times (1-0.6)^{3-3} \times 0.6^3$$

$$= \binom{4-1}{3-1} \times (1-0.6)^{4-3} \times 0.6^3$$

$$= \binom{5-1}{3-1} \times (1-0.6)^{5-3} \times 0.6^3$$

$$= \binom{6-1}{3-1} \times (1-0.6)^{6-3} \times 0.6^3$$

$$= \frac{513}{625}$$

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8) Let X is a random variable with negative geometric distribution of parameters p and r.

a) The probability is:

$$P(X=4) = (1-p)^3 \times p$$
$$= (1-0.09)^3 \times 0.09$$
$$\approx 0.0678$$

b) The probability is:

$$P(X=10) = \binom{10-1}{3-1} \times (1-0.09)^{10-3} \times 0.09^3$$
$$\approx 0.0136$$

c) The expected number is:

$$E(X) = \frac{1}{p} = \frac{1}{0.09} = \frac{100}{9} \approx 11.111$$

d) The expected number is:

$$E(X) = \frac{r}{p} = \frac{3}{0.09} = \frac{100}{3} \approx 33.3333$$

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